

Vicharana Tharkana

Software Engineering Undergraduate | Java & Spring Boot Developer | Full-Stack Web & Android Developer

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Summary

Software Engineering undergraduate with hands-on experience developing full-stack web, Android, and machine learning applications. Skilled in Java, Spring Boot, React, JavaScript, MySQL, and REST API development, with experience implementing authentication, database integration, and recommendation systems. Passionate about building practical software solutions and continuously expanding technical skills.

Education

Open University of Sri Lanka	2021 – Present
Bachelor of Software Engineering (Honours) GPA-3.04	
St. Joseph's College, Bandarawela	2017
Physical Science, Passed with 3S	

Projects

Gemini-Powered E-Commerce Customer Support Chatbot	2025
• Developed an AI-powered chatbot for e-commerce customer support to provide intelligent and automated assistance for users. • Implemented Retrieval-Augmented Generation (RAG) using Astra DB to improve response accuracy using external knowledge sources. • Enabled smart responses for order tracking, shipping updates, and product-related queries with high contextual relevance. • Integrated conversation memory to maintain context across multiple user interactions for a more natural chat experience.	

Tech Stack : Next.js | TypeScript | Google Gemini API | Astra DB
[GitHub](#)

CareerPath Sri Lanka (Full-Stack Web Application)	2025-2026
• Developed a full-stack career guidance platform to help Sri Lankan students discover suitable courses, universities, and career opportunities based on their academic background. • Implemented JWT-based authentication with Spring Security, including role-based access control for Students, Providers, Universities, and Admins. • Designed a smart recommendation system that maps qualification levels and academic streams to relevant career fields and courses.	

Tech Stack : React | Vite | Tailwind CSS | Redux Toolkit | Spring Boot | Spring Security | JWT | MySQL | Spring Data JPA | Swagger/OpenAPI
[GitHub](#)

Explore City Android Application	2023
• Built an interactive mobile application designed to help users explore cities with an intuitive and user-friendly interface.	

- Integrated compass and Google Maps navigation features to provide accurate direction and real-time location guidance.
- Enhanced overall user experience by enabling smooth, location-based exploration and easy access to nearby places.

Tech Stack : Android Studio| Java| Google Maps API| SQLite

[GitHub](#)

Vehicle Care System (Web Application)

2024

- Developed a web-based platform for vehicle maintenance services to streamline service booking and management processes.
- Implemented database integration to efficiently manage service records, user data, and maintenance history.
- Designed user interaction features to improve accessibility and ensure smooth communication between users and service providers.

Tech Stack : HTML| CSS| JavaScript| PHP| MySQL

[GitHub](#)

Chronic Kidney Disease Prediction System

2025

- Developed a neural network-based system to predict Chronic Kidney Disease using patient clinical data.
- Performed data preprocessing, including missing value handling, categorical encoding, and feature scaling.
- Built and trained a TensorFlow/Keras model, achieving **98.75% test accuracy**.
- Designed an interactive prediction interface using Gradio for real-time disease prediction.
- Visualized training accuracy and loss curves for model performance evaluation.

Tech Stack : Python| TensorFlow| Keras| Scikit-learn| Gradio

[GitHub](#)

Skills

Technical Skills

Languages & Frontend : Java, Python, JavaScript, TypeScript, React, Redux Toolkit, Tailwind CSS, HTML/CSS
Backend & Databases : Spring Boot, Spring Security, Express.js, REST APIs, JWT, MySQL, SQLite, MongoDB
Tools & Platforms : Git, GitHub, IntelliJ IDEA, VS Code, Android Studio, Swagger/OpenAPI, Gradio

Soft Skills

Problem Solving, Teamwork, Leadership, Communication, Creativity, Adaptability

Languages

English (Intermediate) , Sinhala (Native)

References

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